VAVE Idea Closure and Lessons Learned Register

FY24 Quarter 1 and Quarter 2

Closed ideas, closure reasons and precedent notes

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Ideas closed in period	337
Distribution	All VAVE idea originators; Product Engineering; Cost Engineering

Summary

Three hundred and thirty-seven ideas were closed in the first half of FY24. One hundred and forty-eight of them, forty-four per cent, were closed as duplicates of ideas already on record. This register exists so that a closure reason can be found and cited when the same proposal returns, rather than being re-analysed from the beginning. Every entry carries the precedent reference needed to close a repeat submission without a fresh engineering assessment.

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1 Purpose and how to use this register

An idea that has been assessed and closed carries information that is expensive to recreate: the load case that defeated it, the supplier constraint, the validation result, or the fact that it was tried and reversed. When that information is not findable, the same idea returns and the same assessment is repeated. This register is the record that makes the second assessment unnecessary.

1.1 Using a precedent

Each closure entry carries a precedent reference. Where a new submission matches an entry in substance, the assessing engineer should close it by citing the precedent and should not repeat the technical assessment. Where the new submission differs in a way that could change the outcome, the difference should be stated explicitly in the closure note.

> Important. A precedent closes a repeat proposal. It does not close a proposal that differs in direction, in scope or in the programme it applies to. A proposal to increase a dimension is not closed by a precedent that rejected reducing it.

2 Closure summary for the period

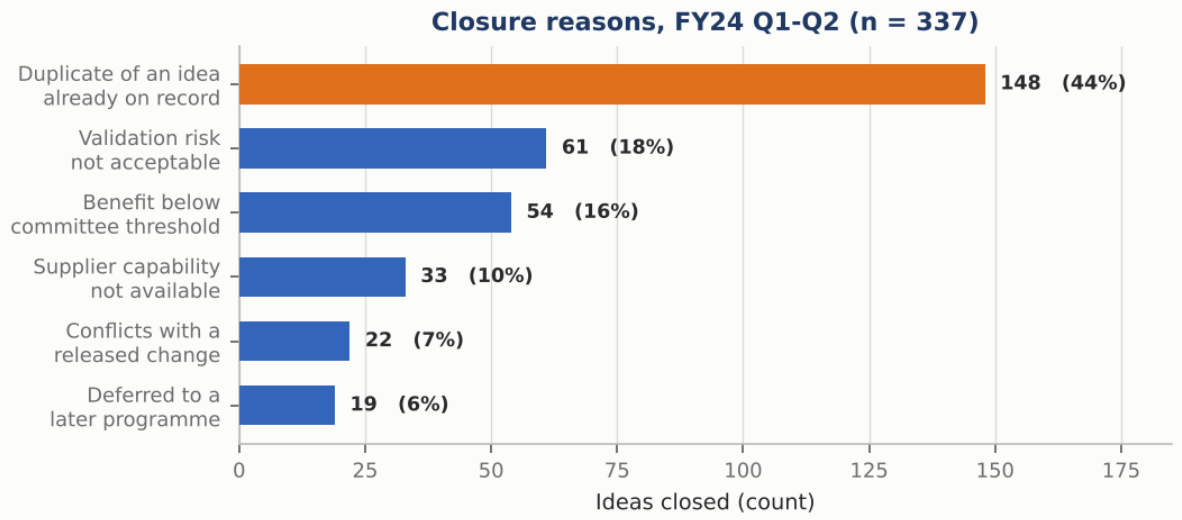


Figure 1 - Closure reasons for the 337 ideas closed in FY24 Q1 and Q2. Duplicates are the largest single category at 44 per cent.

Closure reason	Count	Share	Mean days to close
Duplicate of an idea already on record	148	43.9%	4.2
Validation risk not acceptable	61	18.1%	38.6
Benefit below committee threshold	54	16.0%	12.4
Supplier capability not available	33	9.8%	26.1
Conflicts with a released change	22	6.5%	19.8
Deferred to a later programme	19	5.6%	31.2
Total	337	100%	16.9

Observation 2.1. Duplicates close in 4.2 days on average against 16.9 days for all closures, but only when the precedent is found. Fourteen of the 148 duplicate closures took more than 30 days, in every case because the original idea was located late in the assessment.

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3 Duplicate closures and precedent references

The table below lists the duplicate closures with the highest rate of re-submission. The precedent column gives the idea or document that should be cited when the proposal returns.

Closed ref	Proposal in brief	Component	Precedent	Times seen
VC-24-0041	Steel to aluminium cab mounting bracket	Cab mounting bracket	VC-21-0188, reversed FY24	7
VC-24-0067	Reduce oil pan thickness 1.5 to 1.2 mm	Engine oil pan	VC-22-0310, implemented	6
VC-24-0092	Painted steel to PP battery tray	Battery tray	VC-22-0455, implemented	5
VC-24-0118	Commonise LH and RH mirror brackets	Exterior mirror bracket	VC-21-0402, rejected on shake	5
VC-24-0133	PC to PMMA tail lamp lens	Tail lamp lens	VC-23-0077, implemented	4
VC-24-0156	Narrow front brake drum to 160 mm	Front brake drum	TD-BM/2024/015 finding 5.1	4
VC-24-0171	Delete webs from door hinge bracket	Door hinge bracket	VC-23-0219, implemented	4
VC-24-0208	Commonise front spring brackets	Front spring bracket	TD-BM/2024/015 finding 6.1	3
VC-24-0244	Aluminium exhaust heat shield	Exhaust heat shield	VC-23-0381, implemented	3
VC-24-0290	GFRP mono-leaf front spring	Front spring	CB-BM/2024/007, validation open	3

Observation 3.1. The ten proposals above account for 44 re-submissions in a six-month period. Every one of them has a clear precedent on record. The cost of re-assessment is therefore a retrieval problem, not an engineering one.

3.1 Near-duplicates that were correctly NOT closed

Three submissions in the period superficially matched a precedent but were correctly assessed as distinct. They are recorded here because the distinction is the kind that is easily missed.

Ref	Submission	Apparent precedent	Why it is distinct
VC-24-0101	Increase oil pan thickness 1.2 to 1.5 mm for stone impact	VC-22-0310 reduced it	Opposite direction, different objective
VC-24-0187	Apply released aluminium cab bracket to the tipper programme	VC-21-0188, rigid truck	Different programme and duty cycle
VC-24-0263	PP battery tray plus 40 mm rearward relocation	VC-22-0455, material only	Superset: adds a packaging change

4 Closures on validation risk

Sixty-one ideas were closed because the validation outcome was not acceptable or the validation burden was disproportionate to the benefit. These closures carry the most reusable information, because the validation result itself is the asset.

Closed ref	Proposal	Validation outcome	Status
VC-24-0019	Aluminium cab mounting bracket, 6x4	Mount-face cracking at 62% of target life	Closed, reversed
VC-24-0055	Two-leaf parabolic front spring	Ride acceptance failed at part load	Closed
VC-24-0088	Polymer front hub cap, tipper duty	Cap retention loss after thermal cycling	Closed for tipper only
VC-24-0124	Thinner frame web, 6 to 5 mm, no doubler	Hanger fatigue below target	Closed, superseded by VC-24-0129
VC-24-0129	Thinner frame web with local doubler	Passed at 1.18 times target life	Implemented
VC-24-0203	PMMA lens at high-impact positions	Stone impact failure at 3 of 8 positions	Closed for those positions

Observation 4.1. VC-24-0124 and VC-24-0129 differ only in the addition of a local doubler, and the difference decides the outcome. A closure record that does not capture the doubler would make the successful idea look like a duplicate of the failed one.

5 Closures below the benefit threshold

Fifty-four ideas were closed because the assessed benefit fell below the committee threshold of INR 150 per vehicle. These closures are the most sensitive to input cost movement: an idea closed at INR 120 per vehicle in April 2024 may exceed the threshold at October 2024 material rates.

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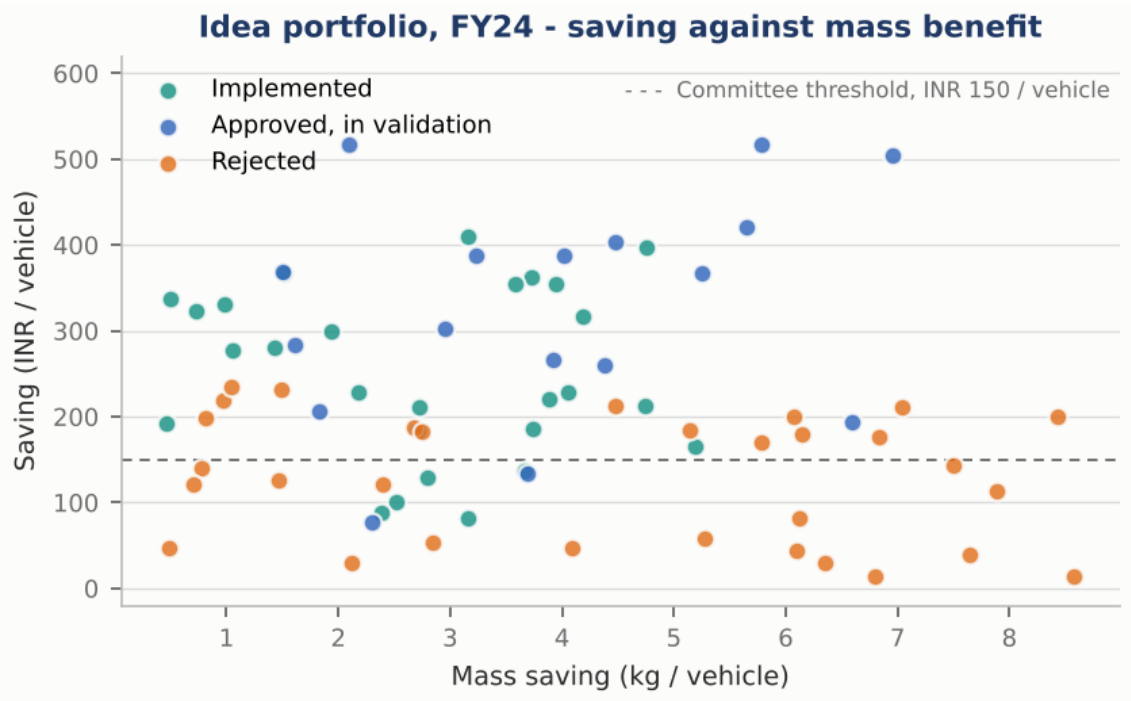


Figure 2 - FY24 idea portfolio by saving and mass benefit. Rejected ideas cluster below the committee threshold but a number sit close to it.

Observation 5.1. Nineteen ideas closed in this period were within INR 30 of the threshold. Against the aluminium and steel movements recorded in CVOEM/VAVE/MC-IX/2025/003, eleven of them would now qualify. A periodic re-test of threshold closures against current material rates is recommended.

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6 Lessons learned

6.1 A reversal is not a failure to record

The aluminium cab mounting bracket was implemented in FY22 and reversed in FY24 after field fatigue cracking. The reversal is recorded, but it is recorded as a separate idea rather than against the original. A search for the original idea therefore returns an implemented change with no indication that it was subsequently withdrawn. Seven re-submissions in this period are attributable to that gap.

> Lesson 6.1. A reversal must be linked to the idea it reverses. An implemented idea that was later withdrawn is more dangerous in the record than one that was never implemented, because it reads as a success.

6.2 Direction is not captured

Three submissions in the period proposed increasing a dimension that an earlier idea had proposed reducing. In two of the three, the assessing engineer initially cited the reducing idea as a precedent. Both were corrected at committee review, but the error is systematic: the two proposals share almost all of their wording and differ only in direction.

6.3 Programme applicability is not captured

Carry-over proposals, in which a change released on one programme is proposed for another, are recorded in the same way as new proposals. They are neither duplicates nor wholly new, and the register has no category for them. Eleven submissions in the period fell into this gap.

> Lesson 6.3. A carry-over proposal should be surfaced against its original, flagged as a carry-over, and assessed only for the difference in duty cycle. Treating it as a duplicate loses a real opportunity; treating it as new repeats work already done.

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7 Repeat submission analysis

Interval since original closure	Repeat submissions	Share
Under 6 months	31	20.9%
6 to 12 months	44	29.7%
12 to 24 months	47	31.8%
Over 24 months	26	17.6%
Total	148	100%

Half of all repeat submissions arrive within twelve months of the original closure, which is well inside the period an experienced engineer would be expected to recall. This indicates that the problem is not memory but discoverability: the originator of a repeat submission is usually not the person who assessed the original.

Originating department	Ideas submitted	Repeats	Repeat rate
Chassis Systems	268	71	26.5%
Cab and Trim	214	58	27.1%
Powertrain Installation	186	41	22.0%
Electrical and Electronics	122	29	23.8%
Manufacturing Engineering	97	18	18.6%
Supplier Development	88	24	27.3%

Observation 7.1. Repeat rates are between 18 and 28 per cent across every department, with no department materially better than the others. A uniform rate across independent groups indicates a systemic retrieval problem rather than a local behavioural one.

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